Principles of Programming Languages Lecture 7: Small-step Structural Operational Semantics

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Outline

Small-step SOS

- States, state operations, rule schemata (in part 1)
- Difference w.r.t. Big-step SOS (in part 1)
- Rules
  - Arithmetic expressions (in part 1)

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Part 1

Recall: states

- The **state** is a *partial finite-domain function*:

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$$\sigma : \mathit{Var} \rightharpoonup \mathit{Int}$$

- Example: $\sigma = x \mapsto 10, y \mapsto 0$

Recall: state operations

- *Lookup*

$$\_(\_) : State \times Var \rightharpoonup Int$$

Recall: state operations

- *Lookup*

$$\_(\_) : \mathit{State} \times \mathit{Var} \rightharpoonup \mathit{Int}$$

- *Update*

$$\_[\_/\_] : \mathit{State} \times \mathit{Int} \times \mathit{Var} \to \mathit{State}$$

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- Since big-step defines all computation steps in one transition it cannot capture non-determinism by default
- Direct control over what to execute and when
- Therefore, small-step SOS is finer-grain than big-step SOS

Small-step SOS

Ingredients:

- *Configuration*
  - Tuple containing semantic ingredients
  - Examples: $\langle a, \sigma \rangle$, $\langle b, \sigma \rangle$, $\langle s, \sigma \rangle$, $\langle p \rangle$, ...

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- *Sequent* (or *transition*)
  - Pair of configurations, to be *derived* or *proved*
  - Examples: $\langle a_1, \sigma\rangle \rightarrow \langle a_1'\rangle$, $\langle a_1 + a_2, \sigma\rangle \rightarrow \langle a_1' + a_2\rangle$, ...
  - Note that we use "$\rightarrow$" instead of "$\Downarrow$" (as for Big-step)

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  - Note that we use "$\rightarrow$" instead of "$\Downarrow$" (as for Big-step)
- *Rule*
  - Tells us how to derive sequents
  - Example:

$$\frac{\langle a_1, \sigma \rangle \rightarrow \langle a_1' \rangle}{\langle a_1 + a_2, \sigma \rangle \rightarrow \langle a_1' + a_2 \rangle}$$

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Rules

Arithmetic expressions (in part 1)

Rule schemata

- Rule schemata:

$$\frac{C_1 \to C_1' \qquad C_2 \to C_2' \qquad \cdots \qquad C_n \to C_n'}{C_0 \to C_0'}$$

Small-step rules for IMP – 1: arithmetic expressions

SMALLSTEP-LOOKUP: $\langle x, \sigma\rangle \rightarrow \langle \sigma(x), \sigma\rangle$ if $\sigma(x) \neq \perp$

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SMALLSTEP-ADD: $\langle i_1 + i_2,\ \sigma\rangle \to \langle i_1 +_{Int} i_2, \sigma\rangle$

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SMALLSTEP-ADD: $\langle i_1 + i_2, \sigma\rangle \to \langle i_1 +_{Int} i_2, \sigma\rangle$

Note: + is non-deterministic!

Simple derivation

- Let us consider $\sigma(x) = 10$. Here is a derivation for sequent $\langle 2 + x, \sigma \rangle \to \langle 2 + 10, \sigma \rangle$:

$$\dfrac{\dfrac{\cdot}{\langle x, \sigma \rangle \to \langle 10, \sigma \rangle}\ \textsc{SmallStep-Lookup}}{\langle 2 + x, \sigma \rangle \to \langle 2 + 10, \sigma \rangle}\ \textsc{SmallStep-Add2}$$

- What about $\langle 2 + x, \sigma \rangle \to \langle 12, \sigma \rangle$?

Sequences

- Reflexive and transitive closure $\rightarrow^*$:

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$$\frac{}{a \rightarrow^* a}\textsc{Refl}$$

Sequences

- Reflexive and transitive closure $\rightarrow^*$:

$$\frac{}{a \rightarrow^* a}\textsc{Refl} \qquad \frac{a_1 \rightarrow a_2 \quad a_2 \rightarrow^* a_3}{a_1 \rightarrow^* a_3}\textsc{Trans}$$

Derivation example

- We can now prove $\langle 2 + x, \sigma\rangle \rightarrow^* \langle 12, \sigma\rangle$:

$$
\dfrac{
  \dfrac{\dfrac{.}{\langle x, \sigma\rangle \rightarrow \langle 10, \sigma\rangle}\ \textsc{Lookup}}{\langle 2 + x, \sigma\rangle \rightarrow \langle 2 + 10, \sigma\rangle}\ \textsc{Add2}
  \quad
  \dfrac{\dfrac{.}{\langle 2 + 10, \sigma\rangle \rightarrow \langle 12, \sigma\rangle}\ \textsc{Add} \quad \dfrac{.}{\langle 12, \sigma\rangle \rightarrow^* \langle 12, \sigma\rangle}\ \textsc{Refl}}{\langle 2 + 10, \sigma\rangle \rightarrow^* \langle 12, \sigma\rangle}\ \textsc{Trans}
}{\langle 2 + x, \sigma\rangle \rightarrow^* \langle 12, \sigma\rangle}\ \textsc{Trans}
$$

Resources

1. Chapter Small-Step Operational Semantics in Software Foundations - Volume 2, Benjamin C. Pierce, Arthur Azevedo de Amorim, Chris Casinghino, Marco Gaboardi, Michael Greenberg, Cătălin Hriţcu, Vilhelm Sjöberg, Andrew Tolmach, Brent Yorgey
`https://softwarefoundations.cis.upenn.edu/plf-current/Smallstep.html`
2. Small-step SOS:
`http://fsl.cs.illinois.edu/images/6/65/CS422-Spring-2015-02c-SmallStep.pdf`